MILITARY SPECIFICATION

TRANSMITTER, SUPERELEVATION: M22

1. SCOPE

1.1 This specification covers one type of electro-mechanical instrument for transmitting superelevation data to a gun elevation system to reduce effect of stereo dip in the optical line of sight (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on data of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–F–13926	—Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.
MIL–F–14619	—Fire Control Parts, Equipment, and Tools; Packaging of.
MIL–I–45208	—Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL–STD–105	—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–130	—Identification Marking of U.S. Military Property.
MIL–STD–202	—Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

ORDNANCE CORPS

F7633152.	—Transmitter, Superelevation: M22.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Initial production sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 **Materials.** Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 **Construction.** The transmitter shall be manufactured in accordance with Drawing F7633152 "Transmitter, Superelevation: M22" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired electrical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.4 **General specifications.** The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926:

(a) Order of precedence
(b) Protective finished surfaces

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

(c) Special accuracy
(d) Finish, painting and protective
(e) Electrical
(f) Workmanship

3.5 Environmental.

3.5.1 *Storage temperatures.* The transmitter shall meet all requirements of this specification at standard ambient temperature (plus 60° to 90° F.) after having been exposed and thermally stabilized at ambient temperatures of plus 160° and minus 80° F.

3.5.2 *Operating temperatures.* The transmitter shall meet the applicable requirements of 3.7 while exposed to and thermally stabilized at ambient temperatures of plus 150° and minus 40° F. Subsequent to which it shall meet the requirements of 3.5.3.

3.5.3 *Vibration.* The transmitter shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of $\frac{1}{16}$-inch ($\frac{1}{8}$-inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration there shall be no loose or missing parts and the transmitter shall meet all requirements of this specification.

3.6 Electrical check.

3.6.1 *Insulation breakdown.* Insulation breakdown shall be in accordance with Specification MIL–F–13926 except that breakdown voltage of 500 volts RMS shall be substituted for 750 volts RMS.

3.6.2 *Insulation resistance.* Insulation resistance shall be in accordance with Specification MIL–F–13926 except that the resistance value shall not be less than 2 megohms and a dc potential of 500 volts shall be used in place of 800 volts.

3.6.3 *Continuity.* The electrical circuits shall be checked for continuity, and shall conform to the wiring diagram.

3.7 Accuracy. With a cable connecting contacts S1, S2, S3, R1 and R2 of a 15TR4A Synchro Torque Transmitter to terminals A, B, C, E and F, respectively, of the input receptacle in the superelevation transmitter, and with a 115 volt, 400 cycle input applied to terminals R1 and R2 of the syncho transmitter, the following requirements shall be met subsequent to vibration when a load of not less than 2 nor more than 8 ounce inches is applied to the output coupling.

Synchro Transmitter rotor rotation (measured in degrees from null position)	*Output coupling rotation (measured in degrees from null position)*
0	0
+ 45	+ 49.3 to + 76.3
+ 90	+ 119.6 to + 146.6
+ 45	+ 64.3 to + 91.3
0	— 6 to + 21
— 45	— 49.3 to — 76.3
— 90	— 119.3 to — 146.6

(Positive values of output shall be considered as clockwise rotation when viewed from the output coupling.)

3.7.1 *Output shaft.* With the 15TR4A Synchro Torque transmitter and the Superelevation Transmitter connected as described in paragraph 3.7, the Synchro Torque Transmitter shall be rotated at a constant speed of 56 plus 2 revolutions per minute for approximately 10 seconds in each direction. The output shaft of the Superelevation Transmitter shall rotate smoothly in each direction and at no time during the 10 second interval shall it stop or reverse direction.

3.8 Slippage, clutch. The clutch shall slip when the motor is energized (with 24 volts direct current plus or minus 0.5 volt) and a load of between 35 and 40 ounce inches is applied to the output coupling.

3.9 Sealing. With the area over the output coupling sealed by placing a cup (with an air inlet) over gasket Drawing A7652186, the instrument shall show no evidence of leakage when subjected to an internal pressure of 2 psi over a period of five minutes. The transmitter shall not be immersed in water to determine area of leakage.

3.10 Identification marking. Identification marking unless otherwise specified shall be in accordance with Standard MIL–STD–130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facili-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

ties and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Inspection requirements and procedures shall be in accordance with Specification MIL–I–45208.

4.3 Initial production sample. Prior to production, the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 *Initial production sample failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation. Inspection lot sizes and lot formation shall be in accordance with Standard MIL–STD–105.

4.4.1 *Defective units and lots.* Defective units and lots shall be processed as specified in Standard MIL–STD–105.

4.5 Material. Control of material shall be in accordance with test data requirements of Specification MIL–I–45208.

4.6 Inspection of components. All positional and functional characteristics shown on drawings, or contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of the superelevation transmitter shall be inspected in accordance with the written inspection system required by Specification MIL–I–45208. These characteristics are defined as requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking and performance.

4.6.1 *Sampling inspection.* Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with Table I.

TABLE I. *Sampling Plan*

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0—25	Use 100% Inspection		
26—35	20	0	1
36—50	25	0	1
51—150	30	0	1
151—300	30	1	2
301—over	60	2	3

4.6.2 *Disposition of nonconforming product.* Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of nonconforming components shall be in accordance with Standard MIL–STD–105.

4.7 Inspection equipment. Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL–I–45208.

4.7.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance ex-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

cept as specified in 4.7.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7.2 *Inspection equipment design requirements.*

4.7.2.1 *Equipment to test accuracy requirements.* The equipment shall consist of a means to measure angular rotation of the output coupling over the range of 360 degrees, accurate to within 2.5 degrees.

4.7.2.2 *Equipment to test slippage requirements.* The equipment shall be capable of applying a load of 35 to 40 ounce-inches to the output coupling, accurate to within 3 ounce-inches.

4.8 **Acceptance inspection.**

4.8.1 *Classification of defects.* The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The AQL specified pertains to each individual characteristic.

4.8.1.1 *Disposition of rejected lots.* Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL–STD–105.

TABLE II. *Classification of Defects*

Use Inspection Level L7 in Table III–B with Sampling Plan Table IV–A of Standard MIL–STD–105

CRITICAL: None defined *MAJOR: AQL 4.0 percent defective*	*Req't. Par.*	*QA. Par.*	*Test Method*
101. Accuracy	3.7	4.9.5	Special Test Equipment (STE)
102. Output shaft	4.7.1	4.9.6	Visual (Vis)
MINOR: None defined			

NOTE: The above tests shall be performed while the superelevation transmitter is stabilized at ambient temperatures of plus 150° and minus 40°.

TABLE III. *Classification of Defects*

Use Inspection Level II in Table III–A with Sampling Plan Table IV–A of Standard MIL–STD–105

MAJOR: AQL 1.5 percent defective	*Req't Par.*	*QA. Pa.*	*Test Method*
101. Physical failure, post vibration	3.5.3	4.9.3	Vis-Tactile
102. Dielectric strength	3.6.1	4.9.4.1	Standard Meas. Equip. (SME)
103. Insulation resistance	3.6.2	4.9.4.2	SME
104. Continuity	3.6.3	4.9.4.3	SME
105. Accuracy	3.7	4.9.5	STE
106. Output shaft	3.7.1	4.9.6	Vis
107. Slippage	3.8	4.9.7	STE
108. Sealing	3.9	4.9.8	SME
109. Identification marking	3.10	4.9.9	Vis.
MINOR: None defined			

NOTE: The above tests shall be performed at standard ambient temperature (plus 60° to plus 90°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

4.9 Test methods and procedures.

4.9.1 *Storage temperatures.* This test shall be conducted in accordance with Procedure II of Specification MIL–F–13926 to the temperatures specified in 3.5.1. Upon return to standard ambient temperature, the superelevation transmitter shall be visually inspected for damage and then subjected to all the tests specified in this specification. This test shall be performed on the initial production sample only.

4.9.2 *Operating temperatures.* This test shall be conducted in accordance with Procedure I of Specification MIL–F–13926 to the temperatures specified in 3.5.2. The superelevation transmitter shall then be subjected to the test of 4.9.5 while thermally stabilized at those temperatures; subsequent to which it shall be vibrated in accordance with 4.9.3.

4.9.3 *Vibration.* Each superelevation transmitter in the sample shall be mounted to a vibrating machine and vibrated in accordance with 3.5.3. Subsequent to vibration the superelevation transmitter shall be subjected to a visual and tactile examination for damage. It then shall be subjected to all the tests of this specification.

4.9.4 *Electrical tests.*

4.9.4.1 *Dielectric strength.* The dielectric strength test shall be performed in accordance with Method 301 of Standard MIL–STD–202 to determine compliance with 3.6.1.

4.9.4.2 *Insulation resistance.* This test shall be performed immediately after the successful completion of the dielectric strength test of 4.9.4.1 in accordance with Method 302, test condition "B", of Standard MIL–STD–202 to determine compliance with 3.6.2.

4.9.4.3 *Continuity.* Continuity shall be tested using standard measuring equipment and the wiring diagram to determine compliance with 3.6.3.

4.9.5 *Accuracy.* The accuracy test shall be accomplished by using a 15 TR4A Synchro Torque Transmitter, connecting terminals S1, S2, S3, R1 and R2 respectively to terminals A, B, C, E and F of the input receptical of the superelevation transmitter. The required voltage shall be applied to terminals R1 and R2 of the synchro transmitter and the specified load applied to the output coupling. When angular rotor rotation of the synchro transmitter is induced in accordance with the values specified in 3.7, the output coupling of the superelevation transmitter shall rotate within the values specified in 3.7. This can be measured on equipment specified in 4.7.2.1.

4.9.6 *Output shaft.* With the synchro torque transmitter connected to the superelevation transmitter as outlined in 4.9.5 and the synchro torque transmitter rotating at the speed, and for the length of time specified in 3.7.1, the output shaft shall be checked visually during rotation for no stoppage or reversal of direction.

4.9.7 *Slippage.* Clutch slippage shall be tested using special test equipment capable of measuring the applied load at the output coupling as specified in 3.8.

4.9.8 *Sealing.* This test shall be accomplished by means of commercial equipment which shall include a sensitive pressure gage capable of testing for the sealing requirements of 3.9.

4.9.9 *Identification marking.* Markings shall be visually inspected for conformance to Standard MIL–STD–130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The superelevation transmitter shall be packaged, packed and marked in accordance with Specification MIL–F–14619/P7633152. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Superelevation Transmitter: M22 is utilized as a component item of the fire control system in the 90 mm Gun Tank: M47. The transmitter receives an electrical signal from the Rangefinder: M12 and actuates a hydraulic pump in the gun

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

elevating system. The pump causes elevation or depression of the 90 mm Gun to reduce to a minimum the difference in angular value, known as stereo dip, between the lines of sight of the range finder, the periscope and the 90 mm Gun when superelevation is fed into the gun elevating system.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for initial production sample including date of delivery (see 3.1).

6.3 **Design information.** Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 **Definitions.**

6.4.1 *Standard measuring equipment (SME).* Standard measuring equipment is defined as the common hand type measuring equipment which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of item under procurement. This category also includes commercial test equipment.

6.4.2 *Special Test Equipment (STE).* Special Testing Equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment. Appearance of "STE" does not imply that the equipment is available, or that a suitable design for such equipment is currently in existence.

6.4.3 *Tactile.* Skillful manipulation or sensing.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Ord

Preparing activity:
Army—Ord

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

FOLD

PARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING OFFICER
FRANKFORD ARSENAL
ATTN: SMUFA-W1100
PHILADELPHIAK, PA. 19137

FOLD

PLATE NO. 15419 (BACK)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:40Z
Check the source to verify that this is the current version before use.